COMS 4720/5720 Principles of Artificial Intelligence

Sample Solution to the Midterm Exam

Fall 2025

1(a) (i) false; (ii) true; (iii) false; (iv) false; (v) true; (vi) true.

(b)

	Breadth-First	Depth-First	Iterative Deepening
Time	$O(b^d)$	$O(b^m)$	$O(b^d)$
Space	$O(b^d)$	$O(bm)$	$O(bd)$

(c) $h_1(n) \geq h_2(n)$ at every node n in the search tree.

(d) 1. m_1.

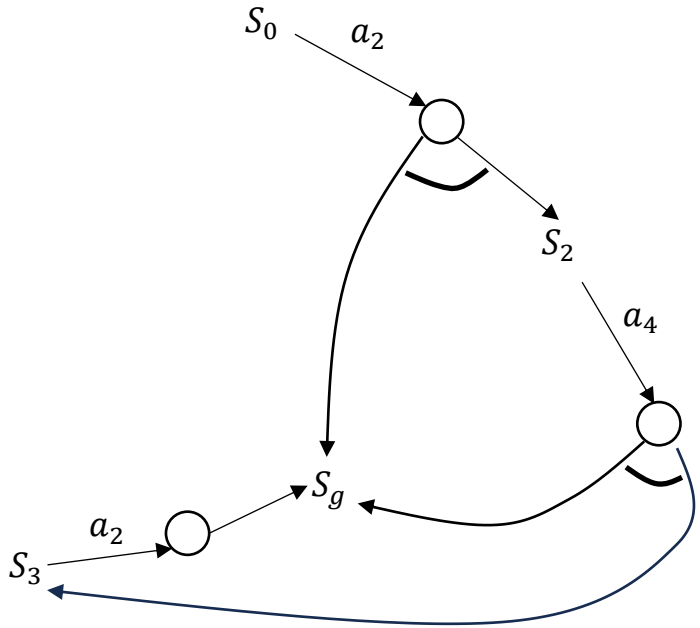
2. MCTS deems the most promising move as the one with the highest number of playouts.

(e) 1. n variables: $X_1, X_2, \ldots, X_n$.

2. $\{r_1, r_2, \ldots, r_n\}$. Same for every variable.

3. either $X_1 < X_2 < \cdots < X_n$ or $X_1 > X_2 > \cdots > X_n$

2.



3(a) A, B, C in the alphabetical order followed by D, E, F in any order.

(b) A: $\{1\}$; B: $\{2\}$; C: $\{1\}$; D: $\{1, 2, 3\}$; E: $\{1, 2, 3\}$; F: $\{1, 2, 3\}$

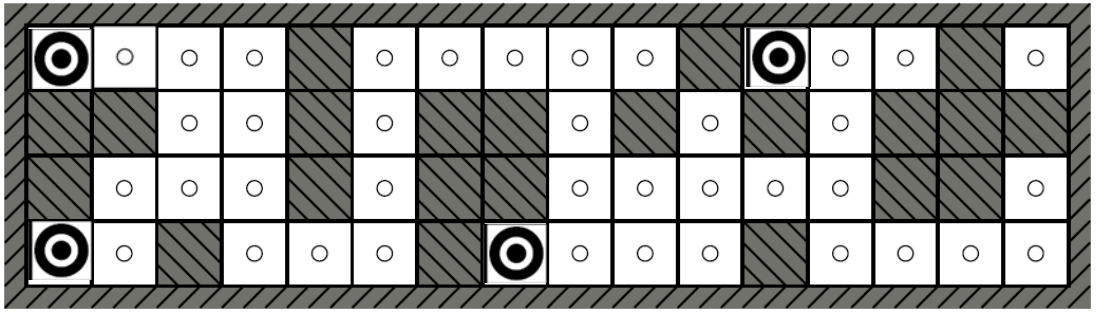
(c) Two solutions: $(1, 2, 1, 1, 1, 3), (1, 2, 1, 1, 2, 3)$. Only E has two feasible values.

4(a) $h_1(S) = 7$ and $h_2(S) = 12$.

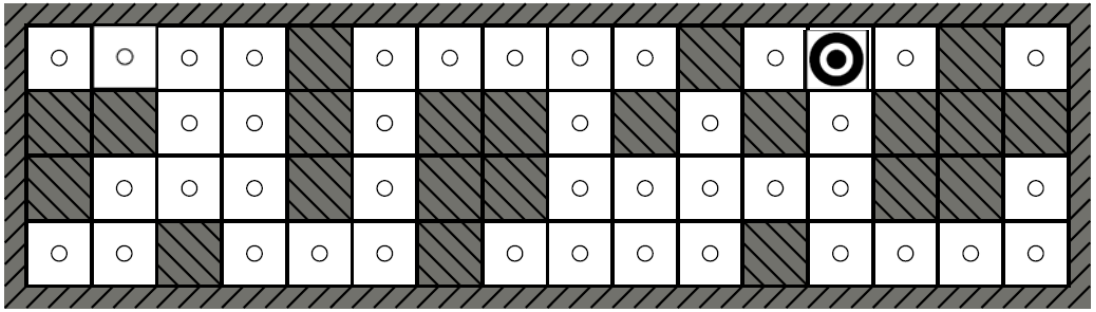
(b) 7; 11.

(c) Yes, because the difference $11 - 7 = 4$ between the two numbers of inversions in (b) is even.

5(a)

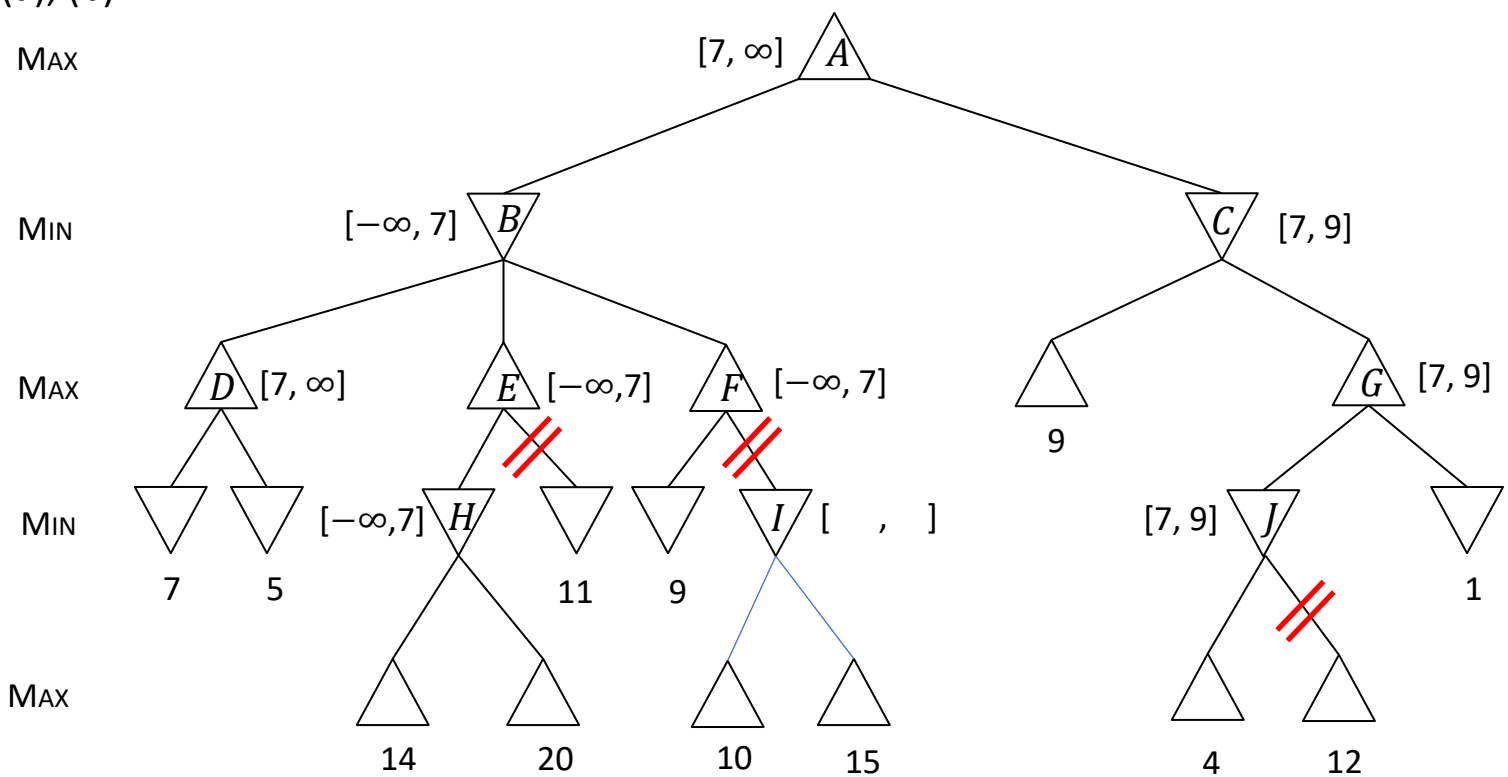


(b)



(c) Yes, complete localization.

6(a), (b).



(c) The final value at the root is 7.